A matrix entry equal to $1+t$: the literal gcd of a nested ribbon commutator

Clio

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Abstract

Let $R_e(t)$ be the height-graded e-ribbon operator on Fock space and, for $e_1,\dots,e_k \geq 2$, let $\mathcal{C}_k = [R_{e_1},[R_{e_2},[\dots,[R_{e_{k-1}},R_{e_k}]\dots]]]$. It is known that $1+t$ divides every matrix entry of $\mathcal{C}_k$ (Q81), and that the $(1+t)$-adic valuation of the gcd of the entries is exactly 1 if and only if $e_{k-1} \neq e_k$ (Q83). The literal statement $\gcd(\mathcal{C}_k) = 1+t$ was left open, because it additionally requires that no other irreducible — in particular neither t nor $t-1$ — divides every entry, and the entire hook family used in Q83 is divisible by t for structural reasons.

We close the gap by exhibiting a *single* matrix entry which is equal to $1+t$ on the nose. Put $E = \sum_i e_i$ and $e_{\min} = \min(e_{k-1},e_k)$. Then

$$\big\langle (E-1,\, e_{\min}) \;\big|\; \mathcal{C}_k \;\big|\; (e_{\min}-1) \big\rangle \;=\; \operatorname{sgn}(e_k - e_{k-1})\,(1+t),$$

for every $k \geq 2$ and all $e_1,\dots,e_k \geq 2$, with no distinctness hypothesis beyond $e_{k-1} \neq e_k$. Consequently $\gcd(\mathcal{C}_k) = 1+t$ literally, in $\mathbb{Z}[t]$ as well as in $\mathbb{Q}[t]$ (the entry is primitive, so no content survives), and Q83's sharpness theorem follows without the closed form for the hook entries.

The proof does not use an involution. The source state $(e_{\min}-1)$ is a one-row partition, chosen so that exactly two beads of its Maya diagram can move, and so that the gap between them equals the displacement of the lower one. The paths then split into two sectors — one in which the beads do not cross, one in which the lower bead overtakes the upper — carrying weights t^0 and t^1 respectively. The illegal configurations of the two sectors are counted by the *same* signed sum X, so the entry is $-X(1+t)$: the factor $1+t$ is the statement that a collision forbidden in one sector is exactly a collision forbidden in the other. A short sign computation gives $X = -\operatorname{sgn}(e_k - e_{k-1})$. All statements are verified against an independent bead-model engine.

1 Setting

Fix an indeterminate t. For a partition λ write $|\lambda\rangle$ for the corresponding basis vector of $\Lambda = \mathbb{Q}(t) \otimes \Lambda_{\mathbb{Z}}$, and let

$$R_e(t)\,|\lambda\rangle = \sum_{\mu} t^{\,h(\mu/\lambda)} |\mu\rangle,$$

the sum over $\mu \supseteq \lambda$ with μ/λ a connected border strip of size e, and h its height (rows minus one).

We use the Maya (bead) model throughout, in the form fixed in `probes/2026-09-04-Q76/abacus.py`: a partition λ is encoded by the bead set

$$B(\lambda) = \{\lambda_i - i \;:\; i \geq 1\} \subseteq \mathbb{Z},$$

a subset of $\mathbb{Z}$ which contains all sufficiently negative integers and no sufficiently large one. Applying R_e means choosing $b \in B$ with $b+e \notin B$ and replacing B by $(B \setminus \{b\}) \cup \{b+e\}$; the weight of that hop is t^h with

$$h = \#\big(B \cap (b, b+e)\big),$$

the number of beads strictly jumped over. We call such a step a *hop of size* e, and we say the hop *crosses* a bead c if $b < c < b+e$. A hop never decreases a bead's position, and beads are indistinguishable only up to the bookkeeping we now fix: we track individual beads, so that a hop moves a particular bead.

Throughout, $e_1, \dots, e_k \geq 2$, $E := \sum_i e_i$, and

$$\mathcal{C}_k = \big[R_{e_1}, \big[R_{e_2}, \big[\dots, [R_{e_{k-1}}, R_{e_k}] \dots\big]\big]\big].$$

We use two results from the earlier papers.

Theorem 1 (Q81, `thm:div`). *$1+t$ divides every matrix entry of $\mathcal{C}_k$, for every $k \geq 2$ and all $e_i \geq 2$, with no distinctness hypothesis.*

Lemma 2 (Q81, `lem:sign`). *Expand $[A_1, [A_2, [\dots, [A_{k-1}, A_k] \dots]]]$ into words. The words that occur are exactly those whose time-ordered index sequence $w = (w_1, \dots, w_k)$ — rightmost operator first, so w_1 is the hop performed first — is unimodal with peak k: $w_1 < \cdots < w_q = k > w_{q+1} > \cdots > w_k$. Each occurs once, with sign $\varepsilon(w) = (-1)^{q-1}$.*

We write $U \subseteq S_k$ for the set of such words. A word $w \in U$ is determined by the set $T \subseteq \{1, \dots, k-1\}$ of letters preceding k:

$$w = \rho_T = \big(T\uparrow,\ k,\ D\downarrow\big), \qquad D := \{1, \dots, k-1\} \setminus T, \qquad \varepsilon(\rho_T) = (-1)^{|T|}, \tag{1}$$

where $T\uparrow$ means the elements of T in increasing order and $D\downarrow$ those of D in decreasing order. In particular

$$\sum_{w \in U} \varepsilon(w) = \sum_{T \subseteq \{1, \dots, k-1\}} (-1)^{|T|} = 0 \qquad (k \geq 2). \tag{2}$$

2 The witness state and its localisation

Standing notation for the rest of the paper. Assume $e_{k-1} \neq e_k$ and put

$$e_{\min} := \min(e_{k-1}, e_k), \qquad p := e_{\min} - 2 \ (\geq 0), \qquad \lambda^\star := (e_{\min} - 1), \qquad \mu^\star := (E-1,\ e_{\min}).$$

Thus $\lambda^\star$ is a single row and $|\mu^\star| = |\lambda^\star| + E$, as required. In beads,

$$B(\lambda^\star) = \mathbb{Z}_{\leq -2} \cup \{p\}, \qquad B(\mu^\star) = \mathbb{Z}_{\leq -3} \cup \{p,\ E-2\}.$$

(Indeed $\lambda^\star_1 - 1 = e_{\min} - 2 = p$ and $\lambda^\star_i - i = -i$ for $i \geq 2$; and $\mu^\star_1 - 1 = E - 2$, $\mu^\star_2 - 2 = e_{\min} - 2 = p$, $\mu^\star_i - i = -i$ for $i \geq 3$.) Note $-2 < p < E - 2$, the last inequality because $E \geq e_{\min} + e_{\max} > e_{\min}$.

Call the bead of $B(\lambda^\star)$ sitting at p the *upper* bead u, and the one at -2 the *lower* bead v.

Lemma 3 (localisation). *Let π be any sequence of k hops, of sizes $e_{w_1}, \dots, e_{w_k}$ in that order, taking $B(\lambda^\star)$ to $B(\mu^\star)$. Then every bead of $B(\lambda^\star)$ at a site ≤ -3 stays put, and no bead ever occupies a site ≤ -3 that it did not start at. Hence only u and v move, and at the end $\{u, v\}$ occupies $\{p, E-2\}$. Exactly one of the following holds.*

(NC) Non-crossing sector: *u ends at $E-2$ and v ends at p. The set $[k]$ of hop labels splits as $S \sqcup \bar{S}$, the hops of u and of v, with $\Sigma_S = E - e_{\min}$ and $\Sigma_{\bar{S}} = e_{\min}$; both parts are nonempty.*

(C) Crossing sector: *u never moves and v ends at $E-2$, performing all k hops.*

Here $\Sigma_A := \sum_{i \in A} e_i$.

Proof. For $y \in \mathbb{Z}$ let $H_y(B) := \#\{x \le y : \ x \notin B\}$ be the number of holes weakly to the left of y; it is finite for every y. A hop moves a bead from some b to $b+e > b$, so it can only turn a site $\le y$ from occupied to empty (if $b \le y < b+e$), never the reverse; hence H_y is non-decreasing along π, and it strictly increases exactly when a bead crosses from $\le y$ to $> y$. Now $H_y(B(\lambda^\star)) = 0 = H_y(B(\mu^\star))$ for every $y \le -3$. Therefore no bead ever moves from a site $\le y$ to a site $> y$, for any $y \le -3$. A bead at $x \le -3$ that hopped would move from x to $x+e > x$ and so cross $y = x \le -3$; so no such bead moves. Since all hops move beads to the right and u, v start at $p \ge 0$ and -2, no bead lands at a site ≤ -3 either.

So only u and v can move, and since $B(\mu^\star)$ differs from $B(\lambda^\star)$ exactly by vacating -2 and occupying $E-2$, the pair $\{u, v\}$ finishes on $\{p, E-2\}$. The bead v starts at $-2 \notin B(\mu^\star)$, so v moves. If v finishes at p then u finishes at $E-2$, giving (NC): the total displacement of v is $p-(-2) = e_{\min}$ and that of u is $E-2-p = E-e_{\min}$, and since the k hops carry total displacement E each hop belongs to exactly one bead, giving the splitting $S \sqcup \bar{S}$; $\bar{S} \neq \varnothing$ as v moves, and $S \neq \varnothing$ because $\Sigma_S = E - e_{\min} > 0$. Otherwise v finishes at $E-2$ and u at p; as u only moves right and p is its start, u never moves, giving (C). $\square$

3 The two sectors

Throughout this section $w \in U$ is a fixed word; w_s is the label of the hop performed at time s.

3.1 The crossing sector

Lemma 4. *In sector (C) every legal path has weight exactly t, and a word $w \in U$ admits a (necessarily unique) path in sector (C) if and only if no partial sum $e_{w_1} + \cdots + e_{w_s}$ $(1 \le s \le k)$ equals $e_{\min}$. Consequently the total contribution of sector (C) to $\langle \mu^\star | \mathcal{C}_k | \lambda^\star \rangle$ is $-t\,X$, where*

$$X := \sum_{\substack{w \in U \\ \exists s:\ e_{w_1}+\cdots+e_{w_s}=e_{\min}}} \varepsilon(w). \tag{3}$$

Proof. In sector (C) the bead v visits $x_0 = -2 < x_1 < \cdots < x_k = E-2$ with $x_s = -2+e_{w_1}+\cdots+e_{w_s}$, so the path is determined by w. Throughout, the other beads sit at p and at $\mathbb{Z}_{\le -3}$; the latter are all $< -2 \le x_{s-1}$ and so are never crossed and never landed on. The hop from x_{s-1} to x_s is legal iff $x_s \neq p$, and its weight is $t^{[x_{s-1}<p<x_s]}$. Since $x_s = p$ is equivalent to $e_{w_1} + \cdots + e_{w_s} = e_{\min}$, legality of the whole path is the stated condition. Given legality, p is not among the x_s, while $x_0 = -2 < p < E-2 = x_k$; so there is exactly one s with $x_{s-1} < p < x_s$, and the total weight is t^1. Summing the signs and using (2),

$$\sum_{\substack{w \in U \\ \text{no partial sum}=e_{\min}}} \varepsilon(w) \cdot t \ = \ t\Big(\sum_{w \in U} \varepsilon(w) - X\Big) \ = \ -tX. \qquad \square$$

3.2 The non-crossing sector

Lemma 5. *Fix a splitting* $[k] = S \sqcup \bar{S}$ *with* $\Sigma_{\bar{S}} = e_{\min}$ *and* $\bar{S} \neq \varnothing$. *In sector (NC), with* $\bar{S}$ *the hops of* v*:*

(a) every hop has weight t^0*;*

(b) the path is legal if and only if $\bar{S}$ *is* not *a time-prefix of* w*, i.e. not equal to* $\{w_1, \dots, w_{|\bar{S}|}\}$.

Consequently sector (NC) contributes $-X'$*, where* $X' := \sum_{\bar{S}} N(\bar{S})$*, the sum over all* $\bar{S} \subseteq [k]$ *with* $\Sigma_{\bar{S}} = e_{\min}$*, and*

$$N(\bar{S}) := \sum_{\substack{w \in U \\ \{w_1,\dots,w_{|\bar{S}|}\} = \bar{S}}} \varepsilon(w). \tag{4}$$

Proof. Write P (resp. Q) for the sum of the sizes of the hops of u (resp. v) performed so far, so that u sits at $p + P$ and v at $-2 + Q$. Since $Q \leq \Sigma_{\bar{S}} = e_{\min}$ we have $-2 + Q \leq p \leq p + P$ at all times: v stays weakly left of p and u weakly right of it.

(a) A hop of u goes from $p + P$ to $p + P + e$, and every site in $(p + P, p + P + e]$ is empty: the beads at $\mathbb{Z}_{\leq -3}$ and the bead v are all $\leq p \leq p + P$. So u's hops are always legal and have weight t^0. A hop of v goes from $-2 + Q$ to $-2 + Q' \leq p$. The only bead that could lie in the open interval $(-2 + Q, -2 + Q')$ is u, at $p + P \geq p \geq -2 + Q'$; so no bead is crossed and the weight is t^0. The hop is legal iff its target is unoccupied, i.e. iff $p + P \neq -2 + Q'$.

(b) By (a), the path is legal iff at each hop of v, say at time s, we have $p + P_{<s} \neq -2 + Q_{\leq s}$, where $P_{<s}$ is the sum of u's hop sizes strictly before time s and $Q_{\leq s}$ that of v's up to and including time s. Since $p = e_{\min} - 2 = \Sigma_{\bar{S}} - 2$, this reads $Q_{\leq s} \neq \Sigma_{\bar{S}} + P_{<s}$. Now $Q_{\leq s} \leq \Sigma_{\bar{S}}$ with equality exactly when s is the last hop of v, and $P_{<s} \geq 0$; so the equality $Q_{\leq s} = \Sigma_{\bar{S}} + P_{<s}$ can hold only when $P_{<s} = 0$ and s is v's last hop — that is, exactly when no hop of u precedes the last hop of v, i.e. exactly when the labels $\bar{S}$ occupy the first $|\bar{S}|$ time slots.

Summing over $w \in U$ for fixed $\bar{S}$ and using (2) gives $\sum_{w \in U} \varepsilon(w) - N(\bar{S}) = -N(\bar{S})$; summing over the admissible $\bar{S}$ gives $-X'$. □

Lemma 6. $X' = X$.

Proof. The partial sums $e_{w_1} + \cdots + e_{w_s}$ are strictly increasing in s because every $e_i \geq 2 > 0$. Hence for a word w there is at most one s with $e_{w_1} + \cdots + e_{w_s} = e_{\min}$, and if there is one, the set $\bar{S} = \{w_1, \dots, w_s\}$ is the unique subset with $\Sigma_{\bar{S}} = e_{\min}$ which is a time-prefix of w. So the words counted in (3) are partitioned according to that $\bar{S}$, and $X = \sum_{\bar{S}} N(\bar{S}) = X'$. □

4 The prefix sign sum

Proposition 7. *Let* $k \geq 2$ *and* $\varnothing \neq \bar{S} \subseteq [k]$*, and let* $N(\bar{S})$ *be as in* (4). *Then*

$$N(\bar{S}) = \begin{cases} (-1)^{|\bar{S}|} & \text{if } k-1 \in \bar{S} \text{ and } k \notin \bar{S}, \\ (-1)^{|\bar{S}|-1} & \text{if } k \in \bar{S} \text{ and } k-1 \notin \bar{S}, \\ 0 & \text{if } \bar{S} \text{ contains both of } k-1, k \text{ or neither.} \end{cases}$$

Proof. Write $r = |\bar{S}|$ and let $w = \rho_T$ as in (1), $m = |T|$, $D = [k-1] \setminus T$. The first r letters of ρ_T are: the r smallest elements of T if $r \le m$; $T \cup \{k\}$ if $r = m+1$; and $T \cup \{k\} \cup \{$the $r-m-1$ largest of $D\}$ if $r > m+1$.

Case $k \notin \bar{S}$. Only $r \le m$ can occur. Then $\{w_1, \dots, w_r\} = \bar{S}$ says: $\bar{S} \subseteq T$ and every element of $T \setminus \bar{S}$ exceeds $\max \bar{S}$. So $T = \bar{S} \sqcup B$ with $B \subseteq (\max \bar{S},\, k-1]$, arbitrary, and $\varepsilon(\rho_T) = (-1)^{r+|B|}$. Hence

$$N(\bar{S}) = (-1)^r \sum_{B \subseteq (\max \bar{S},\, k-1]} (-1)^{|B|} = (-1)^r \cdot \big[\, (\max \bar{S}, k-1] = \varnothing \,\big] = (-1)^r \,[\, \max \bar{S} = k-1 \,],$$

using $\sum_{B \subseteq Y} (-1)^{|B|} = 0$ for $Y \neq \varnothing$. As $k \notin \bar{S}$, $\max \bar{S} = k-1$ iff $k-1 \in \bar{S}$.

Case $k \in \bar{S}$. Put $\bar{S}_0 = \bar{S} \setminus \{k\}$ and $r_0 = r-1$.

If $r = m+1$ then $T = \bar{S}_0$, contributing $(-1)^{r_0}$; this is one term and it always occurs.

If $r > m+1$ then $T \subseteq \bar{S}_0$ with $T \neq \bar{S}_0$, and $\bar{S}_0 \setminus T$ must be the set of the $|\bar{S}_0 \setminus T|$ largest elements of $D = [k-1] \setminus T$. Since $D = ([k-1] \setminus \bar{S}_0) \sqcup (\bar{S}_0 \setminus T)$, this says precisely that every element of $\bar{S}_0 \setminus T$ exceeds every element of $[k-1] \setminus \bar{S}_0$. Let $m^* := \max\big([k-1] \setminus \bar{S}_0\big)$, with $m^* := 0$ if that set is empty, and let $Y := (m^*,\, k-1]$. By the definition of m^* we have $Y \subseteq \bar{S}_0$, and the condition is $\bar{S}_0 \setminus T \subseteq Y$, i.e. $T \supseteq \bar{S}_0 \setminus Y$. Writing $T = (\bar{S}_0 \setminus Y) \cup Y'$ with $Y' \subsetneq Y$, we get $|T| = r_0 - |Y| + |Y'|$ and

$$\sum_{Y' \subsetneq Y} (-1)^{r_0 - |Y| + |Y'|} = (-1)^{r_0 - |Y|} \Big(\sum_{Y' \subseteq Y} (-1)^{|Y'|} - (-1)^{|Y|} \Big) = \begin{cases} -(-1)^{r_0} & Y \neq \varnothing, \\ 0 & Y = \varnothing. \end{cases}$$

Finally $Y = (m^*, k-1]$ is empty iff $m^* = k-1$ iff $k-1 \notin \bar{S}_0$. Adding the two contributions gives $N(\bar{S}) = (-1)^{r_0}\big(1 - [\, k-1 \in \bar{S} \,]\big)$, which is the claim since $r_0 = r-1$. □

Lemma 8. $X = -\operatorname{sgn}(e_k - e_{k-1})$.

Proof. By Lemmas 5 and 6, $X = \sum_{\bar{S}} N(\bar{S})$ over all $\bar{S} \subseteq [k]$ with $\Sigma_{\bar{S}} = e_{\min}$. By Proposition 7 only those $\bar{S}$ containing exactly one of $k-1, k$ contribute.

Suppose $e_{k-1} < e_k$, so $e_{\min} = e_{k-1}$. If $k \in \bar{S}$ then $\Sigma_{\bar{S}} \ge e_k > e_{k-1} = e_{\min}$, impossible. If $k-1 \in \bar{S}$ and $k \notin \bar{S}$ then $\bar{S} = \{k-1\} \cup A$ with $A \subseteq [k-2]$ and $\Sigma_A = e_{\min} - e_{k-1} = 0$, forcing $A = \varnothing$ since every $e_i \ge 2$. So the only contributing set is $\bar{S} = \{k-1\}$, and $X = (-1)^1 = -1 = -\operatorname{sgn}(e_k - e_{k-1})$.

Suppose $e_{k-1} > e_k$, so $e_{\min} = e_k$. Symmetrically, $k-1 \in \bar{S}$ would force $\Sigma_{\bar{S}} \ge e_{k-1} > e_k$; and $k \in \bar{S}$, $k-1 \notin \bar{S}$ forces $\bar{S} = \{k\}$. So $X = (-1)^{1-1} = +1 = -\operatorname{sgn}(e_k - e_{k-1})$. □

5 The theorem

Theorem 9 (the unit witness)**.** *Let $k \ge 2$ and $e_1, \dots, e_k \ge 2$ with $e_{k-1} \neq e_k$. Put $E = \sum_i e_i$ and $e_{\min} = \min(e_{k-1}, e_k)$. Then*

$$\boxed{\big\langle (E-1,\, e_{\min}) \,\big|\, \mathcal{C}_k \,\big|\, (e_{\min} - 1) \big\rangle \;=\; \operatorname{sgn}(e_k - e_{k-1})\,\big(1+t\big).}$$

No distinctness is assumed among $e_1, \dots, e_{k-2}$, and no condition is placed on the position of the largest size.

Proof. By Lemma 3 the entry is the sum of the contributions of sectors (NC) and (C). By Lemma 5 the first is $-X'$, by Lemma 6 $X' = X$, and by Lemma 4 the second is $-tX$. So the entry equals $-X(1+t)$, and Lemma 8 gives $-X = \operatorname{sgn}(e_k - e_{k-1})$. □

Corollary 10 (Q85, the literal gcd)**.** *Under the hypotheses of Theorem 9, the greatest common divisor of the matrix entries of $\mathcal{C}_k$ is $1+t$: in $\mathbb{Q}[t]$ up to a unit, and in $\mathbb{Z}[t]$ up to sign. If $e_{k-1} = e_k$ then $\mathcal{C}_k = 0$ and the gcd is undefined (or 0).*

Proof. By Theorem 1, $1+t$ divides every entry, so $1+t$ divides the gcd g. By Theorem 9 one entry equals $\pm(1+t)$, so g divides $\pm(1+t)$. Hence $g = \pm(1+t)$. The argument is the same in $\mathbb{Z}[t]$ and in $\mathbb{Q}[t]$; in particular no integer content can occur, because $\pm(1+t)$ is primitive. For $e_{k-1} = e_k$ we have $[R_{e_{k-1}}, R_{e_k}] = 0$ and hence $\mathcal{C}_k = 0$. ☐

Corollary 11 (Q83 sharpness, re-proved)**.** *For $k \geq 2$ and $e_i \geq 2$, the $(1+t)$-adic valuation of $\gcd(\mathcal{C}_k)$ is exactly 1 if and only if $e_{k-1} \neq e_k$.*

Proof. Immediate from Corollary 10 and $\mathcal{C}_k = 0$ when $e_{k-1} = e_k$. ☐

Remark 12 (what the mechanism is)**.** The factor $1+t$ is produced here without an involution and without the toggling argument of Q83. It arises because the *same* set of words is forbidden in both sectors: in (NC) a word fails exactly when v's last hop would land on u's seat p before u has vacated it, and in (C) a word fails exactly when v would land on p at all. Lemma 6 says these two exclusion sets have the same signed count X, because a landing on p is the same arithmetic event (a partial sum equal to $e_{\min}$) in both. Since (NC) carries weight t^0 and (C) carries weight t^1, the entry is $-X(1+t)$. In other words: *the collision that blocks the non-crossing sector is the collision that blocks the crossing sector*, and $1+t$ records the two ways the two beads may be arranged around it.

Remark 13 (why the source state had to leave the vacuum)**.** Q83 (§Gaps, item 1) observes that the hook family $\langle (E-j, 1^j)|\mathcal{C}_k|\varnothing\rangle$ cannot witness $t \nmid \gcd$: every such entry equals $\operatorname{sgn}(e_k - e_{k-1})(1+t)\sum_T t^{j-1-|T|}$ with $j-1-|T| \geq 1$, so t divides all of them. That is a property of the hook family, not of $\mathcal{C}_k$; the correct move is to change the *source* state rather than to hunt for a better target. The single row $\lambda^\star = (e_{\min} - 1)$ is the shortest source for which the two-bead gap equals the lower bead's total displacement, which is exactly what makes Lemma 5(b) a prefix condition.

Remark 14 (the $j = e_{\min}$ hook, and a second proof of the corollary)**.** Specialising the Q83 closed form to $j = e_{\min}$ gives a second, independent route. The window $\Sigma_T + e_{\min} \leq j < \Sigma_T + e_{\max}$ at $j = e_{\min}$ admits $T = \varnothing$ and nothing else (a nonempty T needs $\Sigma_T \leq j - e_{\min} = 0$, impossible for $e_i \geq 2$), so

$$\big\langle (E - e_{\min}, 1^{e_{\min}}) \,\big|\, \mathcal{C}_k \,\big|\, \varnothing \big\rangle = \operatorname{sgn}(e_k - e_{k-1})\, t^{e_{\min}-1}(1+t),$$

a monomial times $1+t$, for *every* admissible tuple. Any common divisor of the entries therefore divides $t^{e_{\min}-1}(1+t)$, whose divisors in $\mathbb{Z}[t]$ are $\pm t^i(1+t)^\epsilon$. So Corollary 10 already follows from Theorem 1 together with the weaker statement that *some* entry has nonzero constant term — which is the $t=0$ specialisation of Theorem 9, i.e. sector (NC) alone. We record this because it isolates what is really needed, and because the pair of entries $\big(\pm t^{e_{\min}-1}(1+t),\ \pm(1+t)\big)$ has gcd $1+t$ visibly.

6 Verification

All computations are in the repository `github.com/clio-vega/work-in-progress`, at `work-in-progress@00fd9e` (locally `/home/clio/projects/probes/2026-09-05-Q85/`). The engines they call live at `work-in-progress@00fd` and `:q81/probes/`. Two engines are used, sharing no primitive: the *Maya engine* (`probes/2026-09-04-Q76/abacu` together with `probes/2026-09-04-Q81/nested.py`, which expands the nested bracket into signed words and hops beads on β-sets), and the *two-sector model* (`twobead.py`), which implements Lemmas 4–5 directly as a sum over $(w, \bar{S})$ and never constructs a partition.

1. **Proposition 7** (`twobead.py`, check 1): the closed form for $N(\bar{S})$ was compared with brute-force enumeration over U for every nonempty $\bar{S} \subseteq [k]$ and every $2 \le k \le 7$ — 0 mismatches.

2. **Lemma 5** (`twobead.py`, check 2): the identity $G(\bar{S}) = -N(\bar{S})$, where G is computed by simulating the bead legality condition $Q_{\le s} < g + P_{<s}$ with $g = \Sigma_{\bar{S}}$, was verified for all $\bar{S} \subsetneq [k]$, all $2 \le k \le 6$ and all $\vec{e} \in \{2, \dots, 5\}^k$ — 288,672 cases, 0 mismatches.

3. **Theorem 9** (`verify.py`, checks A and B): the entry computed by the Maya engine was compared with $\operatorname{sgn}(e_k - e_{k-1})(1+t)$ and, separately, with the sector model's prediction $-X(1+t)$, for every tuple with $e_{k-1} \ne e_k$ in $k = 2, 3, 4$ and $e_i \in \{2, \dots, 5\}$, plus a sample at $k = 5, 6$.

4. **Negative control** (`verify.py`, check C): replacing $e_{\min}$ by $e_{\max}$ in the definitions of $\lambda^\star$ and $\mu^\star$ makes the prediction fail on a positive proportion of tuples, so the comparison is not blind.

5. **Corollary 10** (`verify.py`, check D): the literal gcd of the full list of entries of $\mathcal{C}_k$ from the sources $\varnothing$ and $\lambda^\star$ was computed symbolically and equals $1+t$, with integer content 1.

6. **Remark 14** (`phase0.py`): the hook entry at $j = e_{\min}$ was computed from the Maya engine and equals $\operatorname{sgn}(e_k - e_{k-1}) t^{e_{\min}-1}(1+t)$ in every case tested.

7 What remains

1. **The vacuum-restricted gcd.** Corollary 10 is about all matrix entries of $\mathcal{C}_k$, which is the form in which Q83 states its result. The variant "gcd of the entries of $\mathcal{C}_k|\varnothing\rangle$" is *not* proved here. It is true in every case computed (`t0scan.py`: for all $k \le 4$ and $e_i \in \{2, \dots, 5\}$ with $e_{k-1} \ne e_k$, some entry of $\mathcal{C}_k|\varnothing\rangle$ has nonzero constant term), but the witnessing shape varies with $\vec{e}$: for $\vec{e} = (2, 4, 5, 3)$ the two-row shape $(E - e_{\min}, e_{\min}) = (11, 3)$ gives 0 at $t = 0$, and the witnesses are instead $(9, 5)$ and three three-row shapes. From the vacuum the gap between the two top beads is 1, not $\Sigma_{\bar{S}}$, so Lemma 5(b) becomes a genuine ballot condition rather than a prefix condition, and Proposition 7 no longer applies. This is the honest remaining gap.

2. **Sizes $e_i = 1$.** As in Q83, everything is stated for $e_i \ge 2$. The hypothesis is used twice and essentially: in Lemma 6 (strict monotonicity of partial sums, which needs $e_i \ge 1$) and in Lemma 8 ($\Sigma_A = 0 \Rightarrow A = \varnothing$, which needs $e_i \ge 1$). So in fact $e_i \ge 1$ suffices for both, but $p = e_{\min} - 2 \ge 0$ fails at $e_{\min} = 1$ and the localisation lemma would need re-examination. Untested.

3. **The order of N_e (Q84).** Untouched here; Theorem 9 makes it independent of Q83 and Q85 alike.